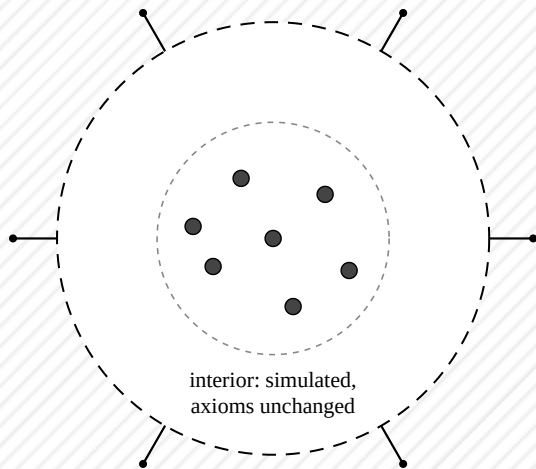


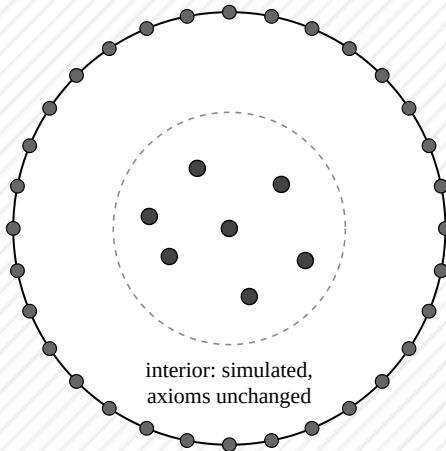
(a) prescribed box (UniverseBox)
nominal radius L $a(t)$ — prescribed



interior: simulated,
axioms unchanged

outside: summarized, not simulated

(b) free box (walls are free masses)
wall = ordinary masses — $a(t)$ is solved



interior: simulated,
axioms unchanged

outside: summarized, not simulated

(c) What every body reads

$$W = D0 + W_B(a) + \sum_j m_j / \sqrt{d^2 + e^2}$$

$D0$ uniform background

W_B box term, $W_B = D a^d - dP$

sum interior sources (simulated)

Semantic commitments (Sec. II A)

1 outside is summarized, not simulated

2 summary = $D0 + W_B +$ (optional grav.)

3 interior physics unchanged

4 box may be prescribed or free

5 comoving coords / wavelength rule

are definitions layered on top

6 no claim about the real universe

Interior coupling to the box

translation: undetectable (exactly)

rotation / expansion: gain $g(r)$,

$g(0) = 1/2$ for a uniform ring (Fig. 2)